PHY4071 Senior Laboratory

Experiment N3: Gamma Ray Spectroscopy

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Abstract

Gamma ray spectroscopy was performed using a NaI(Tl) scintillation detector in combination with both a single-channel analyzer (SCA) and a multi-channel analyzer (MCA). Pulse data was first analyzed to understand how the signal changes throughout the detection system. Spectra were then collected using the SCA by varying the lower level discriminator, and a two-point calibration was used to convert voltage into energy. These spectra were used to identify photopeaks and Compton edges.

Spectra were also obtained using the MCA. A linear calibration equation was used to convert channel number to energy, allowing for identification of photopeaks for various isotopes including ^{54}Mn, ^{60}Co, ^{133}Ba, ^{22}Na, ^{109}Cd, ^{57}Co, and ^{137}Cs. The experimental energies were compared to theoretical values and were generally found to be within uncertainty. Compton edges were also determined and showed reasonable agreement with theoretical predictions.

The full width at half maximum was measured for select peaks and used to calculate the energy resolution of the detector. An unknown isotope was analyzed using its spectra's features and was determined to be ^{137}Cs based on the location of its photopeak and Compton edge.

The results demonstrate that gamma ray spectroscopy using an MCA provides more detailed and accurate spectra compared to the SCA, even though only two isotopes were analyzed via the SCA method, and is an effective method for determining gamma ray energies and identifying radioactive isotopes.

1 Objectives

The objectives of this experiment are:

1. Study the spectra of various gamma-ray emitters using a NaI(TI) scintillation detector.
2. Compare the spectra produced with a single-channel analyzer versus multi-channel analyzer.
3. Perform energy calibration using known sources.
4. Identify an unknown source using its spectrum produced via the multi-channel analyzer.

2 Theory

Gamma-ray spectroscopy is used to analyze the energies of photons emitted from radioactive decay. These photons interact with the detector material and produce signals that can be measured and sorted using a multi-channel analyzer (MCA). By analyzing these signals, the energies of the emitted gamma rays can be determined and used to identify different isotopes.

There are two main interaction processes that contribute to the observed spectra: photoelectric absorption and Compton scattering. Photoelectric absorption is when the gamma photon transfers all of its energy to an electron. This results in a sharp peak in the spectrum, which is the photopeak, and it corresponds directly to the energy of the gamma ray.

Compton scattering is when the gamma photon only transfers a part of its energy to an electron and is scattered with less energy. This produces a continuous distribution of counts
energy in the spectrum, referred to as the Compton continuum. The maximum energy that can be transferred to the electron occurs when the photon is scattered at $180^\circ = \pi$. This point is called the Compton edge. The Compton edge energy is given by the equation

$$E_{CE} = E_\gamma \left(1 - \frac{1}{1 + \frac{2E_\gamma}{m_e c^2}}\right), \tag{1}$$

where E_γ is the energy of the incident photon and $m_e c^2 = 511$ keV.

The MCA records detected events by assigning them to channels based on pulse height, which is proportional to the energy. In order to convert channel number to energy, a calibration must be performed. This is done by identifying known photopeak energies and fitting a linear relationship of the form

$$E = a + bC, \tag{2}$$

where E is the energy, C is the channel number, and a and b are constants. This calibration allows all measured channel values to be converted into physical energy values.

The quality of the detector is described by its energy resolution, which indicates how well it can distinguish between nearby energy peaks. This is determined using the full width at half maximum (FWHM) of a photopeak. The energy resolution is defined as

$$E_{\text{res}} = \frac{\Delta E}{E_\gamma}, \tag{3}$$

where ΔE is the FWHM and E_γ is the energy of the photopeak. This value is expressed as a percentage.

Each isotope produces a unique set of gamma-ray energies and spectra, making gamma spectroscopy an effective method for analyzing radioactive sources.

3 Equipment

3.1 Apparatus List

The principal equipment used in this experiment included:

- High voltage power supply
- Photomultiplier tube
- Pre-amplifier
- Amplifier
- Single channel analyzer (SCA)
- Discriminator
- Scaler
- Multi channel analyzer (MCA)
- UCS-20 software
- Radioactive isotopes with known photopeaks (Table 1) + one unknown isotope

Table 1: Theoretical Gamma-Ray Photopeak Energies

Isotope	**Half-life**	**Expected Gamma Photopeaks (keV)**
^{133}Ba	10.8 years	81, 276, 303, 356, 384
^{109}Cd	462 days	88
^{137}Cs	30.2 years	662
^{57}Co	272 days	122, 136
^{60}Co	5.27 years	1173, 1333
^{54}Mn	313 days	835
^{22}Na	2.6 years	511, 1275

4 Procedures

4.1 Pulse Data Procedure

1. The detection system was assembled using the NaI scintillation detector, photomultiplier tube (PMT), pre-amplifier, and amplifier. The high voltage supply was set to less than 900 V (0.9 kV).

2. The amp gain was adjusted so that the maximum pulse height remained below 10 V to prevent saturation.

3. Pulse shapes were observed at the pre-amplifier output, amplifier output, and SCA output.

4. For each stage of the signal chain, the pulse amplitude, rise time, fall time, and pulse duration were measured from the oscilloscope.

4.2 Procedure for collecting single-channel analyzer data

1. The single-channel analyzer (SCA) and scaler were connected to the detection system after verifying proper pulse shapes.

2. The spectroscopy amplifier shaping time was set to be less than or equal to 2 μs to ensure accurate signal processing.

3. The SCA was set to differential mode, and the ΔE window was adjusted to approximately 0.1 to define the energy range being measured.

4. Gamma sources, specifically ^{60}Co and ^{137}Cs, were used to obtain spectra by varying the SCA threshold setting.

5. For each threshold setting, the count rate was recorded using the scaler, allowing a spectrum of counts versus relative energy to be constructed.

4.3 Procedure for collecting multi-channel analyzer data

1. The SCA and scaler was replaced with the UCS-20 MCA.

2. The following configurations were tested for which produces the best energy resolution:

Table 2: Comparison of MCA Connection Configurations with ^{137}Cs

Parameter	**PA → A**	**PA → PA**	**PA → D**	**A → PA**	**A → A**	**A → D**
Sample	^{137}Cs	^{137}Cs	^{137}Cs	^{137}Cs	^{137}Cs	^{137}Cs
Mode	Pulse height	Pulse height	Pulse height	Pulse height	Pulse height	Pulse height
HV (V)	820	820	820	820	820	820
Coarse gain	1	8	8	8	8	8
Fine gain	2	1.5	2	1	1	1
Conversion gain	1024	1024	1024	1024	1024	1024
LLD (V)	2	2	2	2	2	2
ULD (V)	106	106	106	106	106	106
Amp coarse gain	–	–	–	100	100	100
Amp fine gain	–	–	–	5.6	5.6	5.6

where "PA" is the pre-amp, "A" is the amp, and "D" is direct. It was determined that the Amp-to-Direct configuration produces the best energy resolution. The spectrum was stable with a defined peak and minimal background noise.

3. Calibration was performed with 2-point manual calibration using ^{60}Co (excluding the ^{57}Co sample, which was measured with 3-point manual calibration using ^{60}Co). The following settings were used when collecting data:

Table 3: UCS-20 MCA Two-Point Manual Calibration Settings Using ^{60}Co

Parameter	**Value**
Peak 1 Energy	1173 keV
Peak 2 Energy	1333 keV
Coarse Gain (MCA)	8
Fine Gain (MCA)	1
Amp Coarse Gain	30
Amp Fine Gain	10
Conversion Gain	2048 channels
High Voltage	820 V

5 Data, Analysis, and Results

5.1 Experimental Data Presentation

5.1.1 Pulse data

Table 4: Measured Pulse Characteristics for Different Outputs

Output	**Source**	**Rise Time**	**Fall Time**	**Duration**	**Amplitude**
Pre-amp	Background	1.5 μs	128 μs	129.5 μs	5.6 V
Amp	Background	1400 μs	2800 μs	4200 μs	9.20 V
SCA	^{60}Co	22 ns	198 ns	220 ns	7.0 V

5.1.2 Single-channel analyzer data

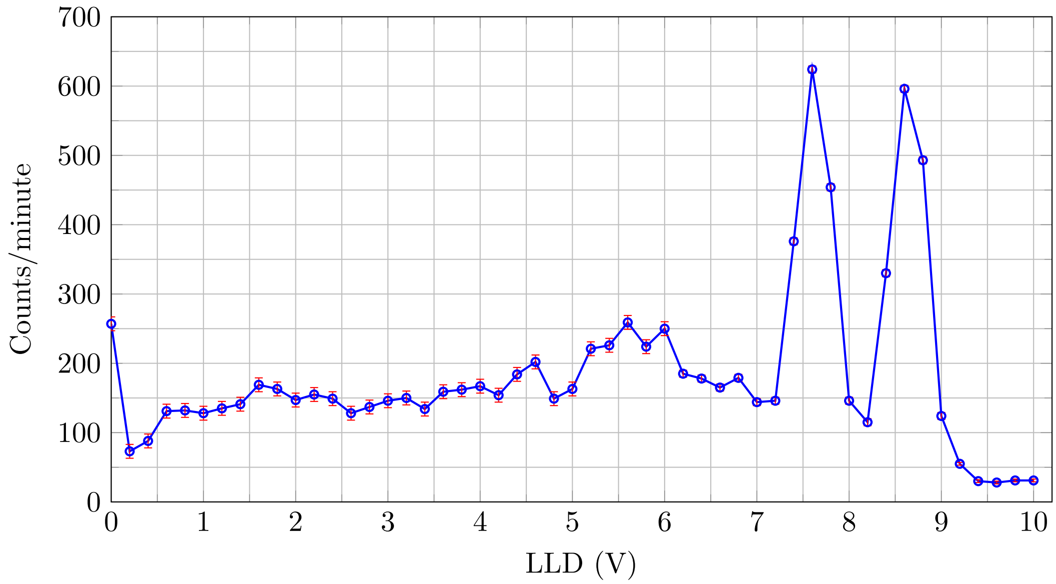

Figure 1: Gamma ray spectrum via SCA for ^{60}Co: Counts per minute as a function of LLD voltage with error.

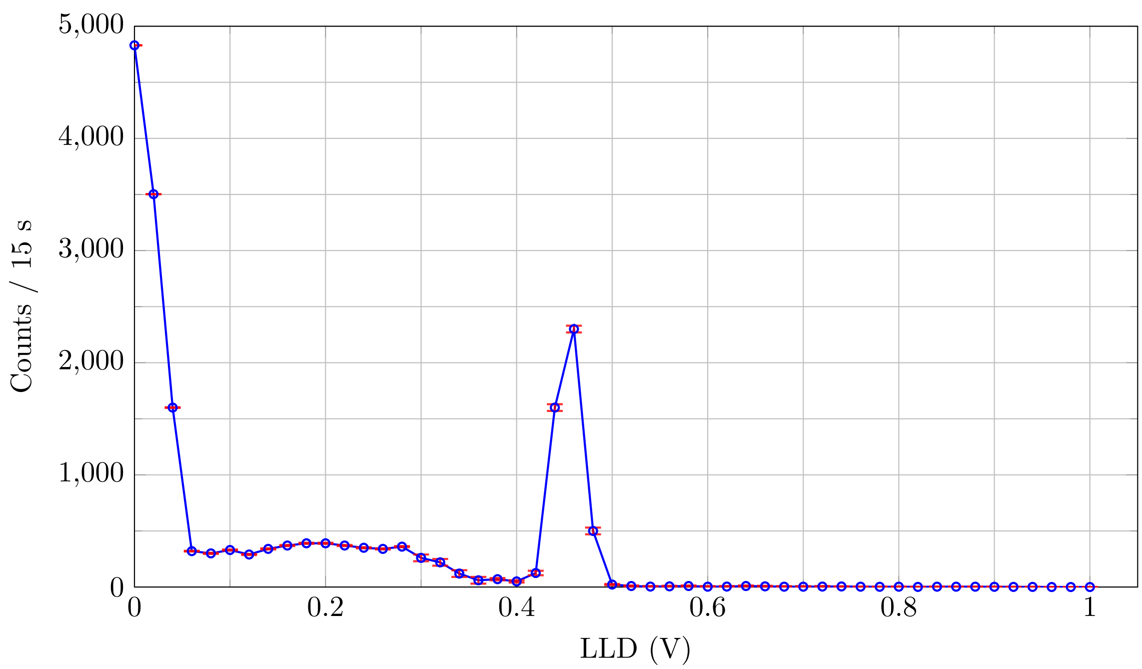

Figure 2: Gamma ray spectrum via SCA for ^{137}Cs: Counts per 15 s as a function of LLD voltage with error.

5.1.3 Multi-channel analyzer data

Each spectra is the result of subtracting the background (Figure 3) from the collected data for the respective isotope.

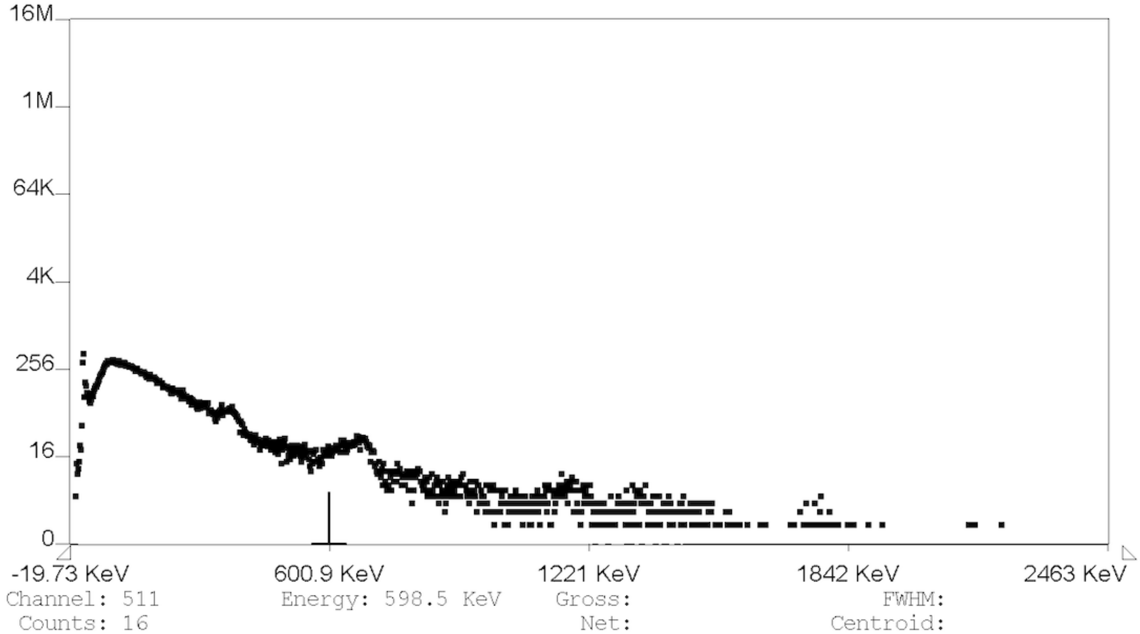

Figure 3: Background spectrum collected with the MCA using Amp-to-Direct configuration.

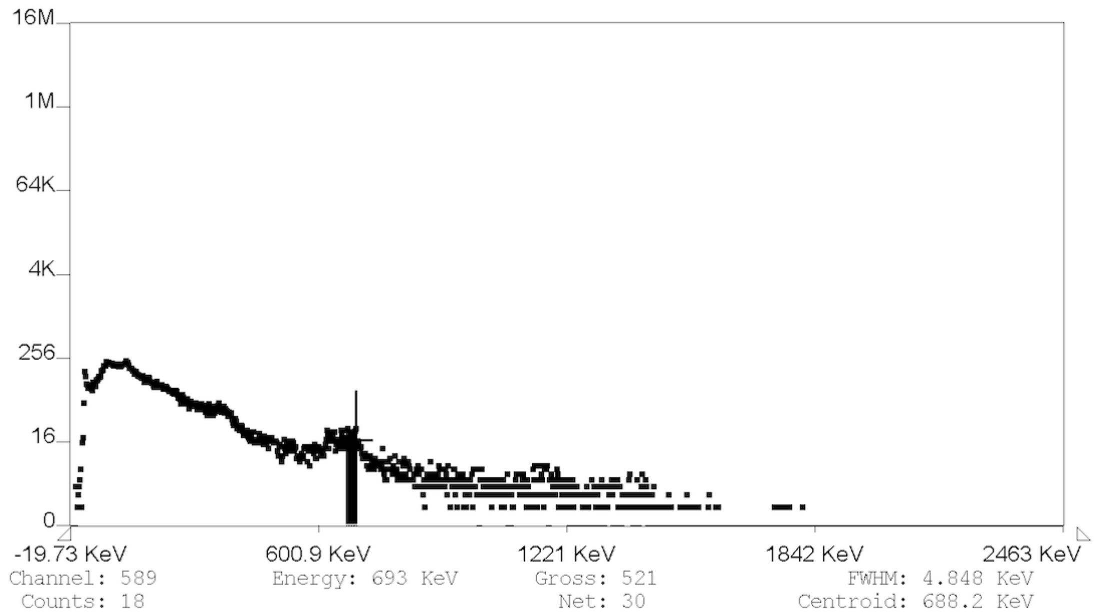

Figure 4: Gamma-ray spectrum of 54Mn collected with the MCA using Amp-to-Direct configuration.

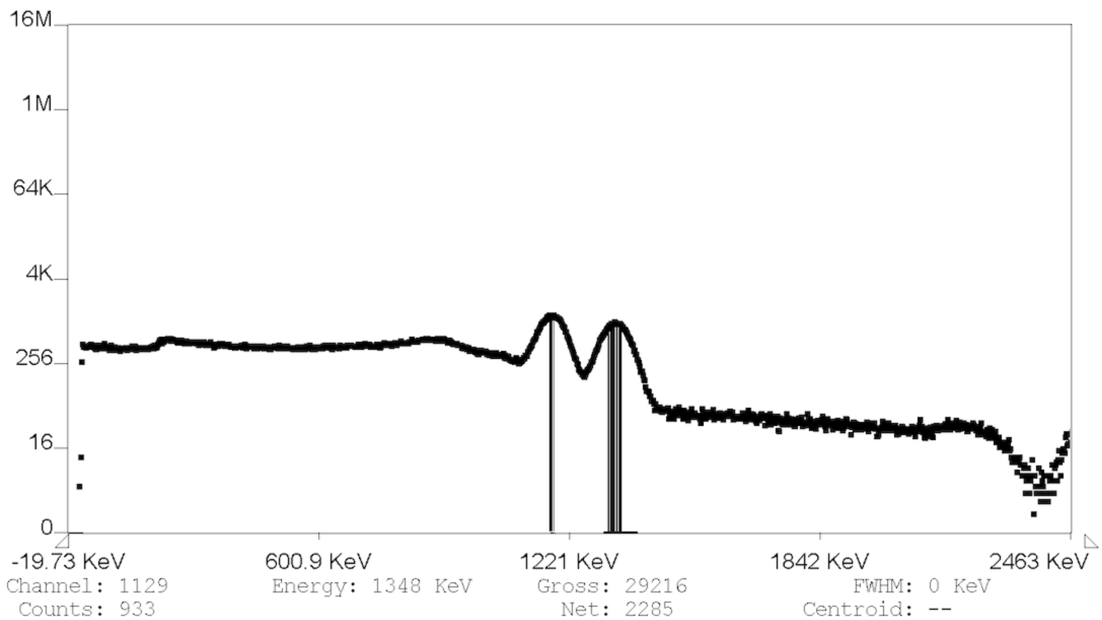

Figure 5: Gamma-ray spectrum of 60Co collected with the MCA using Amp-to-Direct configuration.

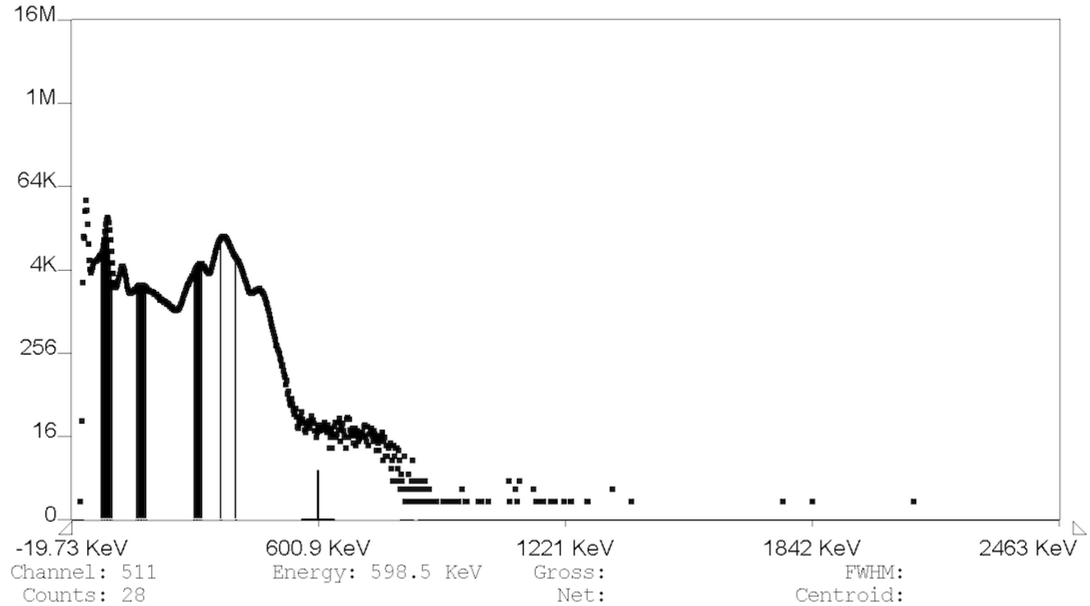

Figure 6: Gamma-ray spectrum of 133Ba collected using the MCA with photopeaks at .

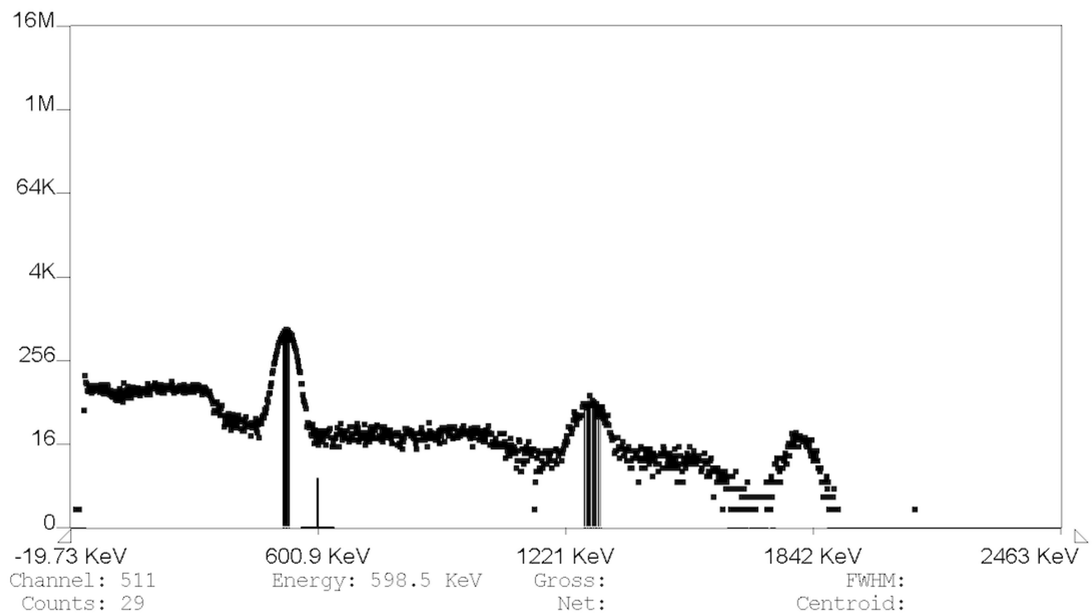

Figure 7: Gamma-ray spectrum of 22Na collected with the MCA using Amp-to-Direct configuration.

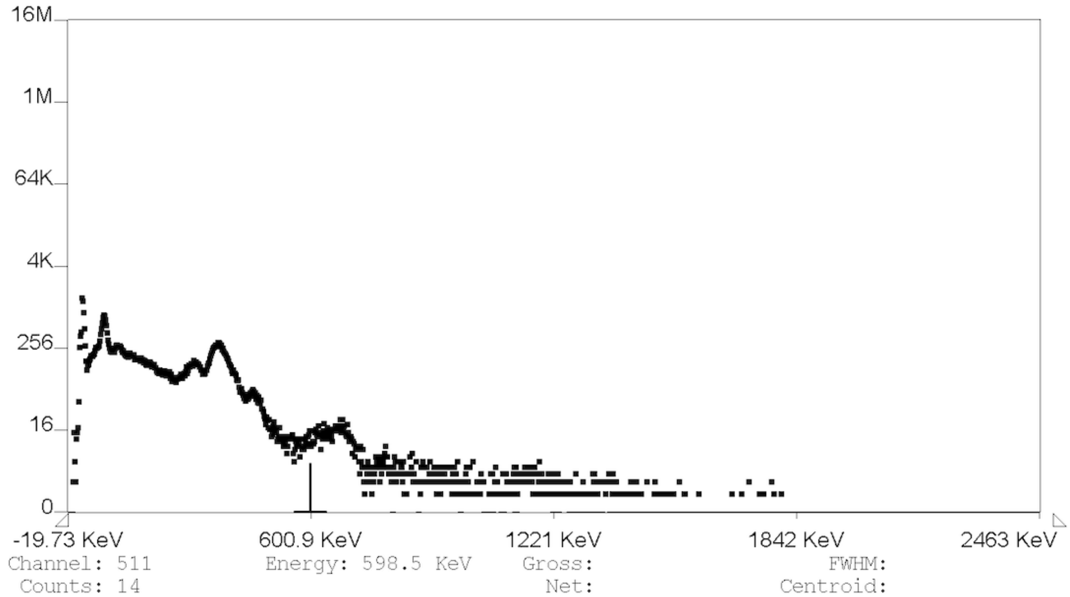

Figure 8: Gamma-ray spectrum of 109Cd collected with the MCA using Amp-to-Direct configuration.

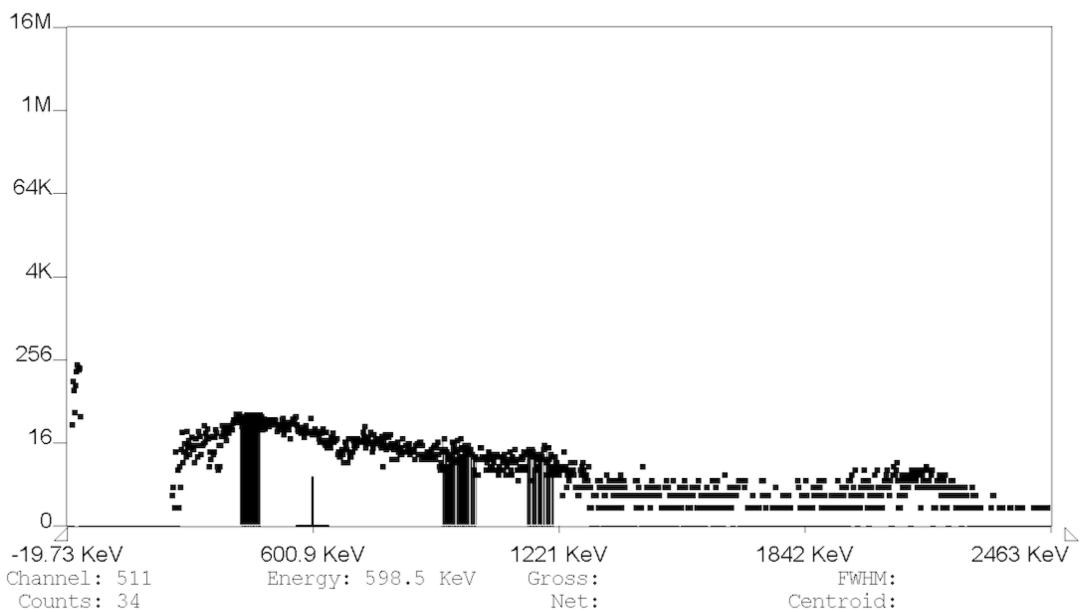

Figure 9: Gamma-ray spectrum of 57Co collected with the MCA using Amp-to-Direct configuration.

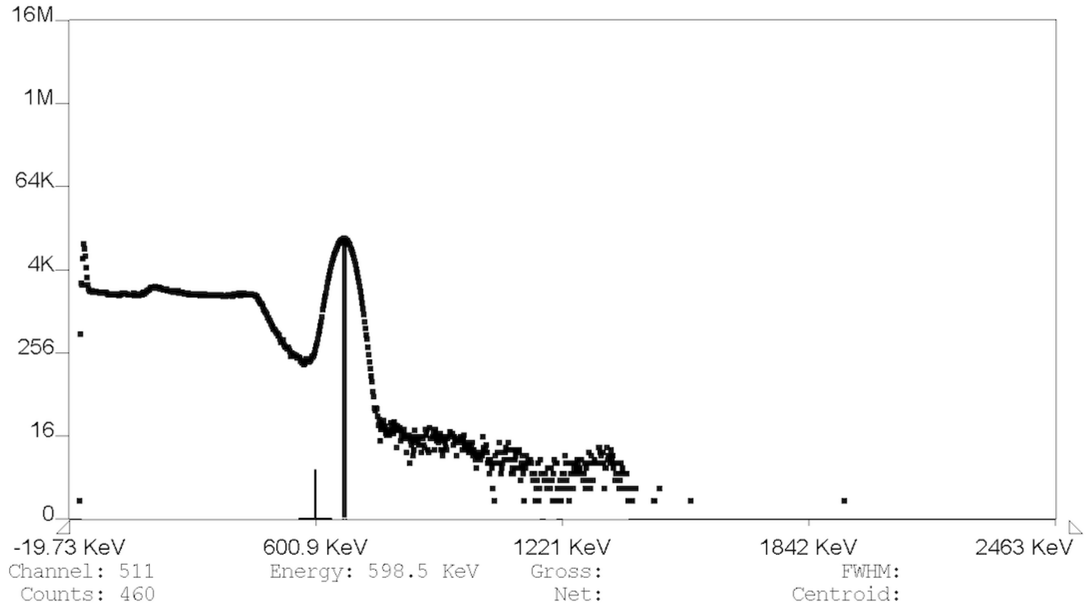

Figure 10: Gamma-ray spectrum of 137Cs collected with the MCA using Amp-to-Direct configuration.

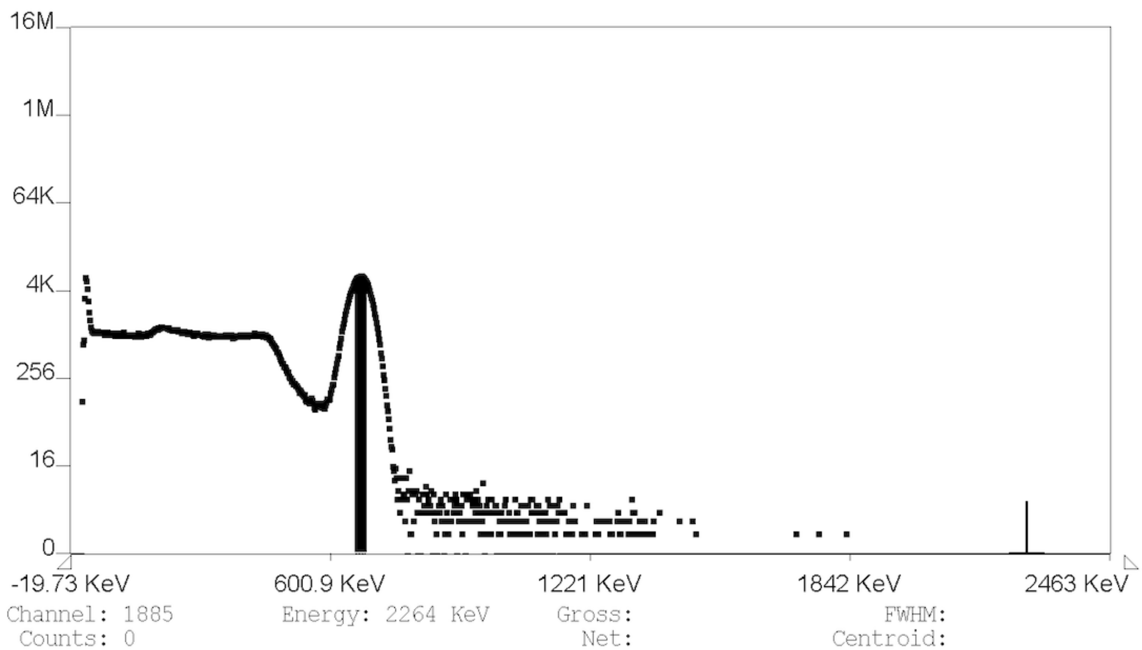

Figure 11: Gamma-ray spectrum of unknown isotope collected with the MCA using Amp-to-Direct configuration.

5.2 Results, Analysis, and Comparison with Theory

5.2.1 Single-channel analyzer

Two-point energy calibration

Two-point energy calibration can be used to convert the detector's arbitrary energy scale into physical energy units. Assume a linear relation between energy E and voltage V:

$$E = mV + b \tag{4}$$

where the slope m is

$$m = \frac{E_2 - E_1}{V_2 - V_1} \tag{5}$$

and the intercept b is

$$b = E_1 - mV_1. \tag{6}$$

The two-point energy calibration is performed using the theoretical photopeaks at 1173 keV and 1333 keV for ^{60}Co. The measured peaks were at 7.6 V and 8.6 V. So, the two-point calibration relation, assuming a linear relationship, is

$$E = 160V - 43 \text{ keV}. \tag{7}$$

This equation can be used to find the energy (in keV) at any point on any spectrum collected using the SCA given its voltage V.

Energy of "third" photopeak of medium energy

There is a structure around $V = 5.8$V on the spectrum of ^{60}Co. Using the two-point calibration relation in Equation (1), it is found that $E = 885$ keV at this structure. ^{60}Co has only two photopeaks at 1173 keV and 1333 keV, so this energy does not correspond to one of its true photopeaks. This structure is likely to be part of the Compton continuum. However, ^{137}Cs only has one true photopeak, so a middle structure cannot be located on its spectrum.

Compton edge energies

The formula for scattered photon energy E' is

$$E' = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2}(1 - \cos\theta)} \tag{8}$$

where E_γ is gamma energy, $m_e c^2$ equals 511 keV, and θ is the scattering angle. The Compton edge corresponds to $\theta = \pi$, where backscatter occurs. Substituting $\theta = \pi$:

$$E_{CE} = \frac{E_\gamma}{1 + \frac{2E_\gamma}{511}}. \tag{9}$$

Using this equation for the first true photopeak of ^{60}Co at 1173 keV yields

$$E_{CE} = \frac{1173 \text{ keV}}{1 + \frac{2(1173 \text{ keV})}{511}} = 963 \text{ keV} = 6.29 \text{ V}. \tag{10}$$

Using Equation (6) again to find the energy at the Compton edge at the second true photopeak of ^{60}Co at 1333 keV:

$$E_{CE} = \frac{1333 \text{ keV}}{1 + \frac{2(1333 \text{ keV})}{511}} = 1118 \text{ keV} = 7.26 \text{ V}. \tag{11}$$

Locating these points on the spectrum for ^{60}Co (Figure 1), the Compton edges can be visually observed around 6.2-6.4 V and 7.2-7.6 V. The theoretical Compton edge energy for the first photopeak of ^{60}Co is 963 keV. The percent difference between the experimental and theoretical Compton edge energies for the first photopeak at 1173 keV is

$$\frac{885 \text{ keV} - 963 \text{ keV}}{963 \text{ keV}}(100) = 0.081 = 8.1\%. \tag{12}$$

The theoretical Compton edge energy for the second photopeak is 1118 keV. The percent difference between the experimental and theoretical Compton edge energies for the second photopeak at 1333 keV is

$$\frac{1125 \text{ keV} - 1118 \text{ keV}}{1118 \text{ keV}}(100) = 0.0063 = 0.63\%. \tag{13}$$

To find the Compton edge for ^{137}Cs, the gamma energy, 662 keV, at the spectrum's only true photopeak is used. Plugging $E_\gamma = 662$ keV and $\theta = \pi$ into Equation (6):

$$E_{CE} = \frac{662 \text{ keV}}{1 + \frac{511}{2(662 \text{ keV})}} = 478 \text{ keV} = 0.35 \text{ V}. \tag{14}$$

The Compton edge for ^{137}Cs can be observed around 0.30-0.36 V on the spectrum for ^{137}Cs (Figure 2). The theoretical Compton edge energy for ^{137}Cs is 478 keV. This is the same as the experimental value, so the percent difference is 0%.

Energy resolution

The equation for energy resolution E_{res} is

$$E_{res} = \frac{\Delta E}{E} = \frac{m\Delta V}{E} \tag{15}$$

where m is the slope from Equation (1) and ΔV is the full width at half maximum (FWHM). The FWHM is calculated by taking the difference of the left and right half maximum voltage. Using $E = 1173$ keV (the first photopeak of ^{60}Co), it is found that $\Delta V = 0.52$ V, so $\Delta E = 160(0.52 \text{ V}) =$ 83 keV. Plugging into Equation (12):

$$E_{res} = \frac{83 \text{ keV}}{1173 \text{ keV}} = 0.071 = 7.1\%. \tag{16}$$

To find the energy resolution for the 662 keV photopeak of ^{137}Cs, we find that $\Delta V = 0.435$

V using the same method as above. Solving for ΔE using the FWHM and slope gives $\Delta E = (160)(0.435) = 62$ keV. Plugging into Equation (12) to solve for the energy resolution:

$$E_{res} = \frac{62 \text{ keV}}{662 \text{ keV}} = 0.094 = 9.4\%. \tag{17}$$

5.2.2 Multi-channel analyzer

The calibration equation acquired from the csv file data from the MCA configuration for each isotope is:

$$E = -19.73 + 2.424C \tag{18}$$

where E is the energy in keV and C is the channel. This equation allows for channel number to be converted into energy, and is consistent for every isotope. The csv file and spectrum for each isotope obtained via the MCA configuration were used to locate the experimental gamma energy at the photopeak(s), which was recorded as the experimental E_{gamma}. The uncertainty in each measurement was calculated using the method shown in this subsection, titled "Error Propagation for MCA Photopeaks". Then the experimental value and theoretical value were compared to determine if the experimental value and its error included the theoretical value. The experimental Compton edge energy can be obtained by plugging the experimental photopeak energy into the Compton edge formula (Equation (6)). The FWHM ΔE can also be calculated around the photopeak region and is used to determine the energy resolution $\frac{\Delta E}{E}$.

Manganese-54

	Th. E_γ (keV)	**Exp. E_γ (keV)**	**Within uncertainty**
Photopeak 1	835	841 ± 24	Yes
Compton Edge 1	640	639	–

Table 5: MCA Spectral Analysis for ^{54}Mn

Cobalt-60

	Th. E_γ (keV)	**Exp. E_γ (keV)**	**Within uncertainty**
Photopeak 1	1173	1173 ± 34	Yes
Photopeak 2	1326	1320 ± 34	Yes
Compton Edge 1	963	963	–
Compton Edge 2	1118	1118	–

Table 6: MCA Spectral Analysis for ^{60}Co

Barium-133

	Th. E_γ (keV)	Exp. E_γ (keV)	Within uncertainty
Photopeak 1	81	82 $\pm$4.4	Yes
Photopeak 2	276	277 $\pm$12	Yes
Photopeak 3	303	310 $\pm$14	Yes
Photopeak 4	356	367 $\pm$15	Yes
Photopeak 5	384	384 $\pm$15	Yes
Compton Edge 1	19.5	20	–
Compton Edge 2	143.5	–	–
Compton Edge 3	164.5	165	–
Compton Edge 4	207.2	–	–
Compton Edge 5	230.4	–	–

Table 7: MCA Spectral Analysis for ^{133}Ba

Sodium-22

	Th. E_γ (keV)	Exp. E_γ (keV)	Within uncertainty
Photopeak 1	511	523 $\pm$17	Yes
Photopeak 2	1275	1270 $\pm$28	Yes
Compton Edge 1	341	335	–
Compton Edge 2	1062	1055	–

Table 8: MCA Spectral Analysis for ^{22}Na

Cadmium-109

	Th. E_γ (keV)	Exp. E_γ (keV)	Within uncertainty
Photopeak 1	88	84 $\pm$4	Yes
Compton Edge 1	22.6	23	–

Table 9: MCA Spectral Analysis for ^{109}Cd

Cobalt-57

	Th. E_γ (keV)	Exp. E_γ (keV)	Within uncertainty
Photopeak 1	122	141 $\pm$8	No
Photopeak 2	136	Not observed	N/A
Compton Edge 1	39	40	–
Compton Edge 2	47	Not observed	–

Table 10: MCA Spectral Analysis for ^{57}Co

Cesium-137

	Th. E_γ (keV)	Exp. E_γ (keV)	Within uncertainty
Photopeak 1	662	671 ±20	Yes
Compton Edge 1	470	478	–

Table 11: MCA Spectral Analysis for ^{137}Cs

Unknown isotope

	Th. E_γ (keV)	Exp. E_γ (keV)	Within uncertainty
Photopeak 1	662	676 ±20	Yes
Compton Edge	478	465	–

Table 12: MCA Spectral Analysis for the unknown isotope, determined to be ^{137}Cs.

FWHM and Energy Resolution

Isotope	Energy (keV)	FWHM (keV)	Energy Resolution (%)
^{60}Co	1173	100	8.6
^{60}Co	1332	103	7.8
^{133}Ba	356	54.6	15.3
^{22}Na	511	48.7	9.5
^{22}Na	1257	105.1	8.2
^{57}Co	122	7.3	7.1
^{137}Cs	662	54.5	8.2

Table 13: FWHM and Energy Resolution calculated for each isotope used in the MCA configuration.

Contamination of ^{109}Cd

^{109}Cd is observed to look very similar to the spectra of ^{133}Ba. Assuming the sample contains ^{133}Ba, the data for ^{109}Cd is analyzed while looking for characteristics of ^{133}Ba.

Th. Energy (keV)	Exp. Energy (keV)	Uncertainty (± keV)	Within Uncertainty
81	65.12	2.92	No
276	276.03	5.90	Yes
303	305.12	6.83	Yes
356	356.03	6.98	Yes
384	382.70	7.34	Yes

Table 14: Barium-133 Photopeak Analysis and Uncertainty Verification

Error Propagation for MCA Photopeaks

The error for the photopeaks in each MCA spectra was calculated as follows:

The width of the peak is also known as it's full-width-maximum-height (FWMH) and is converted into the standard deviation for a Gaussian curve:

$$\sigma = \frac{FWHM}{2\sqrt{2\ln(2)}} \approx \frac{FWHM}{2.355}. \tag{19}$$

The uncertainty in the photopeak is the standard deviation divided by 2. This is because the uncertainty should correspond to the half-width-maximum-height (HWMH) due to defining the uncertainty on either side of the photopeak.

$$\Delta C = \frac{\sigma}{2} \tag{20}$$

Once the uncertainty of the photopeak is located in terms of channels, the uncertainty in terms of energy can be calculated:

$$E_2 = \left(\frac{E_1}{C_1}\right) C_2 \tag{21}$$

$$\Delta E_2 = E_2 \left(\frac{\Delta C_1}{C_1} + \frac{\Delta C_2}{C_2}\right) \tag{22}$$

To show a sample calculation, the energy E_2 is calculated using the ratio of the reference energy ($E_1 = 1173$ keV) to the reference channel ($C_1 = 987$):

$$E_2 = \left(\frac{1173}{987}\right) \times 1069 \approx 1270 \text{ keV} \tag{23}$$

Note: The result is rounded to 3 significant figures, limited by C_1.

The uncertainty ΔE_2 is propagated using the relative uncertainties:

$$\Delta E_2 = 1270 \times \left(\frac{13}{987} + \frac{9.5}{1069}\right) \tag{24}$$

$$\Delta E_2 = 1270 \times (0.013 + 0.0089) \approx 28 \text{ keV} \tag{25}$$

The final result for photopeak energy and its uncertainty is therefore 1270 ± 28 keV.

6 Conclusions

In this experiment, gamma ray spectra were obtained using both a single-channel analyzer (SCA) and a multi-channel analyzer (MCA). Pulse data showed how the signal changes throughout the detection system with different configurations, confirming that the detector was functioning as expected.

The SCA provided a basic method for observing spectral features and allowed for a two-point calibration to convert voltage into energy. However, the MCA produced much more detailed spectra, making it easier to clearly locate photopeaks and Compton edges.

The calibration for the MCA was consistent across all isotopes as two-point calibration using ^{60}Co, expect for ^{57}Co, which was performed with three-point calibration using ^{60}Co, and the experimental photopeak energies were mostly found to be within uncertainty of the theoretical values. Compton edges were also identified and showed agreement with theoretical values. The energy resolution of the detector was determined using the FWHM and was found to vary depending on the energy of the peak, but the calculated values were within reason.

The unknown isotope was identified as ^{137}Cs based on the location of its photopeak and Compton edge. This agrees with the expected spectral characteristics of ^{137}Cs and confirms the accuracy of the identification.

Overall, the experiment demonstrates that gamma ray spectroscopy using an MCA is an effective method for measuring gamma ray energies and identifying radioactive isotopes.

References

[1] Adrian C. Melissinos, *Experiments in Modern Physics*. Academic Press, 2nd edition, 1966.

[2] R. S. Jin and M. Hohlmann, *Experiment N-3: Gamma Ray Spectrometry*, 2024. Senior Physics Laboratory Manual.

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[5] ORTEC, *Introduction to Preamplifiers*, 2020. AMETEK ORTEC technical note.

[6] ORTEC, *Introduction to Amplifiers*, 2020. AMETEK ORTEC technical note.

[7] S. M. Lee et al., *Gamma-ray spectroscopy*. arXiv, 2024.

A Raw Data

Table 15: SCA Count Rate as a Function of Lower Level Discriminator (LLD) for ^{60}Co

LLD (V)	**Counts/minute**
0.0	257
0.2	73
0.4	88
0.6	131
0.8	132
1.0	128
1.2	135
1.4	141
1.6	169
1.8	163
2.0	147
2.2	155
2.4	149
2.6	128
2.8	137
3.0	146
3.2	150
3.4	134
3.6	159
3.8	162
4.0	167
4.2	154
4.4	184
4.6	202
4.8	149
5.0	163
5.2	221
5.4	226
5.6	259
5.8	224
6.0	250
6.2	185
6.4	178
6.6	165
6.8	179
7.0	144
7.2	146
7.4	376
7.6	624
7.8	454
8.0	146
8.2	115
8.4	330
8.6	596
8.8	493
9.0	124
9.2	55
9.4	30
9.6	28
9.8	31
10.0	31

Table 16: SCA Count Rate as a Function of Lower Level Discriminator (LLD) for ^{137}Cs

LLD (V)	**Counts / 15 s**
0.00	48
0.02	350
0.04	1600
0.06	320
0.08	300
0.10	330
0.12	290
0.14	340
0.16	370
0.18	390
0.20	390
0.22	370
0.24	350
0.26	340
0.28	360
0.30	260
0.32	220
0.34	120
0.36	60
0.38	70
0.40	50
0.42	125
0.44	1600
0.46	2300
0.48	500
0.50	22
0.52	9
0.54	4
0.56	6
0.58	9
0.60	3
0.62	4
0.64	9
0.66	6
0.68	3
0.70	3
0.72	4
0.74	5
0.76	3
0.78	2
0.80	3
0.82	1
0.84	3
0.86	3
0.88	3
0.90	2
0.92	1
0.94	1
0.96	0
0.98	0
1.00	0